

PCA and SVD

Exercise 1

In this exercise, you will use the Fisher Iris dataset to perform Principal Component Analysis (PCA) via Singular Value Decomposition (SVD). The dataset comprises 150 samples across three species (setosa, versicolor, virginica), with four features each (sepal length, sepal width, petal length, petal width). You are provided with the MATLAB file:

`PCA_SVD_IRIS_ex.m`

The file aims to implement PCA using SVD.

- (a) Read through the file and fill in the missing parts.
- (b) Discuss the results.
- (extra-c) Compare the results with Matlab's own PCA function.